

# DBMS PRACTICE 2

## Question1 Create Table

```
mysql> create table student_marks1 (  
-> roll_no int primary key,  
-> name varchar(50),  
-> subject varchar(50),  
-> marks decimal(5,2)  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> show tables;
```

```
+-----+  
| Tables_in_task2 |  
+-----+  
| student_marks1 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> desc student_marks1;
```

| Field   | Type         | Null | Key | Default | Extra |
|---------|--------------|------|-----|---------|-------|
| roll_no | int          | NO   | PRI | NULL    |       |
| name    | varchar(50)  | YES  |     | NULL    |       |
| subject | varchar(50)  | YES  |     | NULL    |       |
| marks   | decimal(5,2) | YES  |     | NULL    |       |

4 rows in set (0.00 sec)

## Question2. Insert Data

```
mysql> insert into student_marks1 (roll_no, name, subject, marks) values  
-> (1, 'ravi', 'math', 85.50),  
-> (2, 'sita', 'math', 92.75),  
-> (3, 'anil', 'math', 78.40),  
-> (4, 'priya', 'math', 88.90),  
-> (5, 'vijay', 'math', 80.25),  
-> (6, 'subbusir', 'aws', 98.50),  
-> (7, 'dwaraka', 'dbms', 95.75),  
-> (8, 'ranjani', 'english', 97.40),  
-> (9, 'kavithamam', 'aws1', 99.90),  
-> (10, 'seetha', 'azure', 82.25);
```

Query OK, 10 rows affected (0.01 sec)

Records: 10 Duplicates: 0 Warnings: 0

## Question3. Simple Select

```
mysql> select * from student_marks1;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 1       | ravi       | math    | 85.50 |
| 2       | sita       | math    | 92.75 |
| 3       | anil       | math    | 78.40 |
| 4       | priya      | math    | 88.90 |
| 5       | vijay      | math    | 80.25 |
| 6       | subbusir   | aws     | 98.50 |
| 7       | dwaraka    | dbms    | 95.75 |
| 8       | ranjani    | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha     | azure   | 82.25 |

```
10 rows in set (0.00 sec)

mysql>
```

#### Question4. Aggregate Functions

```
mysql> select count(*) as total_students from student_marks1;
```

| total_students |
|----------------|
| 10             |

```
1 row in set (0.01 sec)

mysql> select sum(marks) from student_marks1;
```

| sum(marks) |
|------------|
| 899.60     |

```
1 row in set (0.01 sec)

mysql> select avg(marks) from student_marks1;
```

| avg(marks) |
|------------|
| 89.960000  |

```
1 row in set (0.00 sec)

mysql> select max(marks) from student_marks1;
```

| max(marks) |
|------------|
| 99.90      |

```
1 row in set (0.01 sec)

mysql> select min(marks) from student_marks1;
```

| min(marks) |
|------------|
| 78.40      |

```
1 row in set (0.00 sec)

mysql> |
```

#### 5. Filtering with WHERE

## Question

```
mysql> select * from student_marks1  
-> where marks > 85;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 1       | ravi       | math    | 85.50 |
| 2       | sita       | math    | 92.75 |
| 4       | priya      | math    | 88.90 |
| 6       | subbusir   | aws     | 98.50 |
| 7       | dwaraka    | dbms    | 95.75 |
| 8       | ranjani    | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |

7 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> where marks >= 90;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 2       | sita       | math    | 92.75 |
| 6       | subbusir   | aws     | 98.50 |
| 7       | dwaraka    | dbms    | 95.75 |
| 8       | ranjani    | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |

5 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> where marks < 80;
```

| roll_no | name | subject | marks |
|---------|------|---------|-------|
| 3       | anil | math    | 78.40 |

1 row in set (0.00 sec)

```
mysql> select * from student_marks1  
-> where marks between 80 and 90;
```

| roll_no | name   | subject | marks |
|---------|--------|---------|-------|
| 1       | ravi   | math    | 85.50 |
| 4       | priya  | math    | 88.90 |
| 5       | vijay  | math    | 80.25 |
| 10      | seetha | azure   | 82.25 |

4 rows in set (0.01 sec)

```
mysql> select * from student_marks1  
-> where name like 'p%';
```

| roll_no | name  | subject | marks |
|---------|-------|---------|-------|
| 4       | priya | math    | 88.90 |

1 row in set (0.00 sec)

```
mysql> select * from student_marks1  
-> where name in ('ravi', 'sita', 'vijay');
```

| roll_no | name  | subject | marks |
|---------|-------|---------|-------|
| 1       | ravi  | math    | 85.50 |
| 2       | sita  | math    | 92.75 |
| 5       | vijay | math    | 80.25 |

3 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> where marks > 85  
-> and (subject = 'math' or name like 'p%');
```

| roll_no | name  | subject | marks |
|---------|-------|---------|-------|
| 1       | ravi  | math    | 85.50 |
| 2       | sita  | math    | 92.75 |
| 4       | priya | math    | 88.90 |

3 rows in set (0.00 sec)

```
mysql> |
```

## Question

### 6. Update Queries

```
mysql> update student_marks1
      -> set marks = 90.00
      -> where roll_no = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from student_marks1
      -> where roll_no = 3;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      3 | anil | math    | 90.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> |
```

```
mysql> update student_marks1
      -> set subject = 'remedial math'
      -> where marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql> select * from student_marks1
      -> where marks < 80;
Empty set (0.00 sec)
```

```
mysql> |
```

## Question7. Delete Queries

## Question

```
mysql> delete from student_marks1
      -> where roll_no = 5;
Query OK, 1 row affected (0.01 sec)

mysql> select * from student_marks1;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 1       | ravi       | math    | 85.50 |
| 2       | sita       | math    | 92.75 |
| 3       | anil       | math    | 90.00 |
| 4       | priya      | math    | 88.90 |
| 6       | subbusir   | aws     | 98.50 |
| 7       | dwaraka    | dbms    | 95.75 |
| 8       | ranjani    | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha     | azure   | 82.25 |

```
9 rows in set (0.00 sec)

mysql> delete from student_marks1
      -> where marks < 75;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from student_marks1;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 1       | ravi       | math    | 85.50 |
| 2       | sita       | math    | 92.75 |
| 3       | anil       | math    | 90.00 |
| 4       | priya      | math    | 88.90 |
| 6       | subbusir   | aws     | 98.50 |
| 7       | dwaraka    | dbms    | 95.75 |
| 8       | ranjani    | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha     | azure   | 82.25 |

```
9 rows in set (0.00 sec)
```

## 8. Sorting

## Question

```
mysql> select * from student_marks1  
-> order by marks asc;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 10      | seetha     | azure   | 82.25 |
| 1       | ravi       | math    | 85.50 |
| 4       | priya      | math    | 88.90 |
| 3       | anil       | math    | 90.00 |
| 2       | sita       | math    | 92.75 |
| 7       | dwaraka    | dbms    | 95.75 |
| 8       | ranjani    | english | 97.40 |
| 6       | subbusir   | aws     | 98.50 |
| 9       | kavithamam | aws1    | 99.90 |

9 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> order by marks desc;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 9       | kavithamam | aws1    | 99.90 |
| 6       | subbusir   | aws     | 98.50 |
| 8       | ranjani    | english | 97.40 |
| 7       | dwaraka    | dbms    | 95.75 |
| 2       | sita       | math    | 92.75 |
| 3       | anil       | math    | 90.00 |
| 4       | priya      | math    | 88.90 |
| 1       | ravi       | math    | 85.50 |
| 10      | seetha     | azure   | 82.25 |

```
mysql> select * from student_marks1  
-> order by name asc;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 3       | anil       | math    | 90.00 |
| 7       | dwaraka    | dbms    | 95.75 |
| 9       | kavithamam | aws1    | 99.90 |
| 4       | priya      | math    | 88.90 |
| 8       | ranjani    | english | 97.40 |
| 1       | ravi       | math    | 85.50 |
| 10      | seetha     | azure   | 82.25 |
| 2       | sita       | math    | 92.75 |
| 6       | subbusir   | aws     | 98.50 |

9 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> order by name desc;
```

| roll_no | name       | subject | marks |
|---------|------------|---------|-------|
| 6       | subbusir   | aws     | 98.50 |
| 2       | sita       | math    | 92.75 |
| 10      | seetha     | azure   | 82.25 |
| 1       | ravi       | math    | 85.50 |
| 8       | ranjani    | english | 97.40 |
| 4       | priya      | math    | 88.90 |
| 9       | kavithamam | aws1    | 99.90 |
| 7       | dwaraka    | dbms    | 95.75 |
| 3       | anil       | math    | 90.00 |

9 rows in set (0.00 sec)

```
mysql> |
```

## Question

### 9. Group By

```
mysql> select subject, sum(marks) as total_marks
-> from student_marks1
-> group by subject;
```

| subject | total_marks |
|---------|-------------|
| math    | 357.15      |
| aws     | 98.50       |
| dbms    | 95.75       |
| english | 97.40       |
| aws1    | 99.90       |
| azure   | 82.25       |

6 rows in set (0.02 sec)

```
mysql> select subject, avg(marks) as avg_marks
-> from student_marks1
-> group by subject;
```

| subject | avg_marks |
|---------|-----------|
| math    | 89.287500 |
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
| azure   | 82.250000 |

6 rows in set (0.00 sec)

```
mysql> select subject, count(*) as num_students
-> from student_marks1
-> group by subject;
```

| subject | num_students |
|---------|--------------|
| math    | 4            |
| aws     | 1            |
| dbms    | 1            |
| english | 1            |
| aws1    | 1            |
| azure   | 1            |

6 rows in set (0.00 sec)

```
mysql> |
```

## Question10. Having

```
mysql> select subject, avg(marks) as avg_marks
-> from student_marks1
-> group by subject
-> having avg(marks) > 90;
```

| subject | avg_marks |
|---------|-----------|
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |

4 rows in set (0.00 sec)

```
mysql> select subject, count(*) as num_students
-> from student_marks1
-> group by subject
-> having count(*) > 1;
```

| subject | num_students |
|---------|--------------|
| math    | 4            |

1 row in set (0.00 sec)

```
mysql> |
```

Question